PART: Baking Attributes into High-Dimensional Vectors for Filtered Approximate Nearest Neighbor Search

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ABSTRACT

Filtered Approximate Nearest Neighbor Search (FANNS), which retrieves similar high-dimensional vectors subject to attribute constraints, is critical for numerous applications. However, existing FANNS methods suffer substantial search performance degradation under low pass rate conditions, meaning only a tiny fraction of the dataset satisfies the constraints. This is mainly because prior methods cannot unify attributes and vectors in a shared space: prioritizing attributes induces disconnected sub-indexes, hindering efficient search traversal, whereas prioritizing vectors prevents valid candidates from clustering, forcing the algorithm to exhaust its search budget on invalid candidates. Motivated by this, we propose PART, which natively bakes attributes into the vector space via an attribute-aware rotation over raw vectors. This rotation preserves proximity among vectors with identical attributes while pushing vectors of different attributes toward near-orthogonality. PART provides theoretical error bounds to guarantee accurate distance computation on these attribute-baked vectors. Furthermore, we extend PART to support complex predicates, including multi-attribute AND, numeric range filters, and hybrid workloads. Finally, PART is integrated into mainstream ANN indexes, including RaBitQ quantization, IVF, and graph indexes. Comprehensive experiments validate that PART achieves a favorable Pareto frontier between search efficiency and accuracy across all datasets. Under an extreme low pass rate of 0.1% and a target recall of 0.95, PART achieves up to a 5.5× speedup over the state-of-the-art baseline.

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PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://github.com/Lsyhprum/PART.

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1 INTRODUCTION

Unstructured data objects such as images, text, and videos are typically mapped to *high-dimensional vectors* encoding their semantic content. These vectors are then queried using Approximate Nearest Neighbor Search (ANNS) to return the top-k objects similar to a given query [23, 34, 44, 53]. While ANNS has been widely applied in content retrieval [16, 18], recommendation systems [6, 56], and Retrieval-Augmented Generation (RAG) [26, 37], modern applications increasingly rely on structured *attribute* constraints to enhance the quality of semantic query results. For example, an enterprise RAG system may retrieve relevant design images belonging to a specific department, or an AI agent might search for similar financial reports published within a specific time window. Such queries require retrieving the top-k objects that are most semantically similar (e.g., in visual or textual proximity) among those satisfying attribute constraints (e.g., departmental categories or temporal bounds). To meet this practical demand, Filtered Approximate Nearest Neighbor Search (FANNS) has emerged as an important problem and attracted growing attention [1, 4, 29, 37, 54].

Limitations of Existing Methods. Many FANNS methods have been proposed with the aim of pushing the Pareto frontier of search efficiency and accuracy [27, 48]. A key obstacle lies in handling variable *pass rate*, the fraction of vectors satisfying filter conditions [22]. Early paradigms including Pre-filtering and Post-filtering rely on attribute-vector decoupled execution strategies [46]: Pre-filtering resolves the attribute constraints before vector search, but often degenerates into an expensive brute-force scan under high pass rates [40]. Conversely, Post-filtering retrieves candidates first and discards invalid matches later, wasting massive distance computations when valid candidates are sparse under low pass rates [27]. These paradigms fail to evaluate attributes and vectors jointly, rendering them highly fragile across a wide spectrum of pass rates.

Recently, Joint-filtering has established itself as the mainstream FANNS paradigm. This paradigm is typically built upon graph-based indexes [34, 49], achieving FANNS either by restructuring *attribute-dedicated* indexes during the construction phase (e.g., FDiskANN [13], iRangeGraph [52]), or by fusing attribute matching into the routing logic of an *attribute-agnostic* index during the search phase (e.g., ACORN [40], NaviX [41]). However, Joint-filtering struggles to maintain the robust connectivity crucial for efficient graph traversal under low pass rates. This is fundamentally

Table 1: Comparison of PART against existing Joint-filtering methods. More ★'s indicate better.

	Ours (PART)	Attribute-dedicated (e.g., FDiskANN)	Attribute-agnostic (e.g., ACORN)
Attribute Format	A sequence of random phase angles.	Implicit in graph topology.	External structured metadata.
Filtering Strategy	Transforms vectors via phase-based rotation.	Restructures index topology.	Attribute-aware routing strategy.
Predicate Support	Common predicates (`EQ`, `AND`, range, hybrid).	Single predicate.	Arbitrary predicates.
Index Compatibility	Highly compatible with VQ, IVF, and graph.	Requires designing custom structures.	Requires modifying search algorithms.
Pass Rate Adaptability	Preserves nearest-neighbors proximity and attribute distinctiveness. ★★★	Fragmented topology lacking cross-attribute proximity edges. ★	Topologically disconnected candidates waste distance computations. ★

because these methods still treat the attribute space and vector space as distinct entities, failing to natively unify them. Consequently, for attribute-dedicated methods, the partitioned sub-indexes struggle to reliably establish proximity edges across different attribute values. Conversely, for attribute-agnostic methods, the attribute matched candidates remain topologically disconnected, forcing the search to waste massive distance computations on invalid neighbors.

Our Solution PART. To overcome the pass rate limitations of existing methods by breaking this fundamental structural tradeoff, we propose PART, a Phase-Based Attribute Rotation Transform method that natively bakes structured attributes into a continuous vector space while simultaneously maintaining precise evaluations of vector similarity and attribute matching with theoretical guarantees. Its key insight is encoding attribute values into distinct random *phase angles*, which are subsequently utilized to drive an *attribute-aware rotation* on the vector space. Vectors sharing identical attribute values undergo the same angular transformation to preserve their original spatial proximity, thereby ensuring the accuracy of vector similarity results. Conversely, vectors with differing attribute values are separated by a sufficiently large angular margin, guaranteeing the soundness of attribute matching. By unifying attributes and vectors at the representation level, PART allows mainstream ANN indexes to inherently support FANNS without requiring ad-hoc modifications to their core graph topology or routing logic.

PART achieves this transformation through a two-step pipeline: **(1)** PART constructs a high-dimensional composite phase angle vector for each attribute value, where each element in the angle vector represents a 2D coordinate plane. This is achieved by repeatedly evaluating a dedicated-seed random hash function across the $B = d/2$ disjoint dimension pairs. **(2)** PART translates each angle from the angle vector into a 2×2 standard rotation block for its respective dimension pair, assembling them into a d-dimensional block-diagonal orthogonal matrix. By applying this matrix, the attribute value is baked directly into the data vector. The orthogonal nature of this transformation preserves the similarity between vectors sharing identical attribute values. Furthermore, by leveraging extensive theoretical tools related to the Johnson-Lindenstrauss Lemma [9, 39], we prove that the similarity between vectors with mismatched attribute values tightly concentrates around zero.

We further extend the basic PART to support complex predicates, including multi-attribute `AND`, numeric ranges, and hybrid workloads that combine discrete attributes with numeric ranges. Specifically, for *multi-attribute* `AND`, it superimposes the phase rotations of all associated attributes. For *numeric ranges*, it anchors the baseline rotation at the range center, translating numerical deviations into proportional angular shifts to guarantee distance preservation for valid targets and attenuation for boundary violations. Ultimately, for *hybrid workloads*, PART seamlessly unifies discrete and numeric filters within the same space, leveraging its highly compositional phase rotation framework.

Finally, because this transformation operates entirely at the data representation level, PART serves as a highly compatible module. We integrate it with mainstream ANNS methods, including RaBitQ [12], IVF, and graph-based indexes, by operating directly on the transformed vectors without altering their core search logic or topological structures. Furthermore, by deterministically parameterizing the transformation of each attribute value with its own dedicated pseudo-random seed, PART incurs *negligible storage overhead* and rapidly reconstructs the rotation matrices on-the-fly.

We evaluate PART extensively on standard and real-world datasets, empirically validating three core advantages. **(1)** Compared with state-of-the-art baselines, PART achieves up to 5.5× higher QPS at a pass rate of 0.1%, and maintains a 1.5× to 3× advantage at a high pass rate of 50%. Furthermore, it sustains up to a 2.5× speedup on ArXiv dataset, which features a highly *skewed* real-world attribute distribution. **(2)** PART robustly maintains a strict 90% Recall@100 target even under highly constrained filtering spaces, such as an extreme 0.001% pass rate or ultra-large attribute cardinalities of up to 10,000. **(3)** Our empirical results on real-world datasets align tightly with our established theoretical bounds, rigorously confirming the reliability and completeness of our theoretical analysis.

Contributions. We summarize our contributions as follows:

- We identify that mainstream Joint-filtering FANNS methods fail to natively unify structured attribute and continuous vector spaces, inherently causing connectivity bottlenecks that severely degrade search performance under low pass rates.
- We propose PART, a novel method that bakes attributes into a continuous vector space, enabling accurate evaluation of attribute matching and vector similarity through a unified computation.
- We extend PART to support complex predicates, including multi-attribute `AND`, numeric ranges, and hybrid workloads.
- We provide theoretical guarantees establishing the attribute matching soundness and vector similarity completeness for both single-attribute filtering and multi-attribute `AND`. Additionally, we derive error bounds for numeric ranges and hybrid filtering.

2 PRELIMINARIES

We formalize Approximate Nearest Neighbor Search (ANNS) and its filtered variant FANNS, which restricts search to vectors satisfying attribute constraints. Table 2 summarizes frequently used notation.

2.1 Approximate Nearest Neighbor Search

Given a dataset $\mathcal{D} \subset \mathbb{R}^d$ of N data vectors in a d-dimensional space and a query vector $\mathbf{q}_r \in \mathbb{R}^d$, Nearest Neighbor Search (NNS) returns the ground-truth set $\mathcal{G}(\mathbf{q}_r) \subset \mathcal{D}$ of the k data vectors

Table 2: Frequently used notation.

Notation	Definition
$\mathcal{D}$	the vector dataset
$\mathbf{o}_r$, $\mathbf{q}_r$	the raw data and query vector
$\mathbf{o}$, $\mathbf{q}$	the normalized data and query vector
$\mathcal{A}$	the attribute schema
$\mathbf{a}$, $\mathbf{t}$	the attribute set of data vector and query target
a, t	the attribute value and query target value
$\mathbf{p}_a$	the phase angle vector of attribute value a
$\hat{\mathbf{o}}$, $\hat{\mathbf{q}}$	the attribute-baked data vector and query vector

most similar to $\mathbf{q}_r$ under a given distance metric. Since exact search incurs prohibitive computational cost at scale, Approximate Nearest Neighbor Search (ANNS) is widely used in practice. ANNS retrieves a candidate set $\mathcal{R}(\mathbf{q}_r)$ of k approximate nearest neighbors, trading a minor drop in search quality for substantially higher efficiency. ANNS quality is measured by Recall, which is formally defined as:

$$\text{Recall@}k(\mathbf{q}_r) = \frac{|\mathcal{G}(\mathbf{q}_r) \cap \mathcal{R}(\mathbf{q}_r)|}{k} \quad , \tag{1}$$

while the search efficiency of ANNS is typically measured by query throughput, i.e., Queries Per Second (QPS).

ANNS methods are commonly classified by their underlying structure into four families: tree- [7, 32, 36], hash- [14, 19, 28], IVF- [3, 5, 24], and graph-based methods [10, 11, 34]. The latter two dominate industrial deployments [2, 38, 47]. Both approaches evaluate distances on full-precision vectors, which constitutes the main computational cost at large scale. To mitigate this cost, Vector Quantization (VQ) is widely coupled with IVF and graphs for faster search and smaller memory footprint. For example, FAISS [23] uses coarse clusters to prune the search space, relying on PQ [24] to accelerate distance computations within the target clusters. DiskANN [44] compresses vectors into PQ codes to fit in memory, fetching raw vectors from an SSD only for reranking. SymphonyQG [15] embeds low-bit quantizers [12] into the graph traversal process, thereby bypassing the overhead of full-precision distance evaluations.

2.2 Filtered ANNS

In modern real-world applications, vectors are widely accompanied by structured metadata. Suppose that each raw data vector $\mathbf{o}_r \in \mathcal{D}$ is associated with attributes $\mathbf{a}$ (e.g., $\mathbf{a} = \{\text{color} : \text{“blue”},\ \text{price} : 85\}$) drawn from a schema $\mathcal{A}$. Filtered Approximate Nearest Neighbor Search (FANNS) addresses retrieval over such data by formulating a query as a pair $(\mathbf{q}_r, \phi)$, coupling the raw query vector $\mathbf{q}_r$ with a boolean predicate ϕ over $\mathcal{A}$ (e.g., color = “blue” `AND` price < 100). This predicate defines the target subset of eligible candidates $\mathcal{D}_\phi = \{\mathbf{o}_r \in \mathcal{D} \mid \phi(\mathbf{a}) = \text{true}\}$. The fraction of these eligible candidates is defined as the pass rate of predicate ϕ, denoted as:

$$\text{pass rate} = \frac{|\mathcal{D}_\phi|}{|\mathcal{D}|} \quad . \tag{2}$$

When $|\mathcal{D}_\phi| \approx k$ or smaller, the search must recover nearly all eligible candidates, making the problem hardest. Since pass rates drastically impact query performance [27], numerous FANNS methods have been proposed, which can be broadly categorized into three paradigms based on the execution order of vector search and attribute filtering: Pre-filtering [51], Post-filtering, and Joint-filtering. Pre-filtering first materializes the matched attributes as a bitmap or ID-list, followed by a brute-force vector search over that subset. While straightforward, it *suffers severely under high pass rates* [27, 31], where the subset of eligible candidates $\mathcal{D}_\phi$ approaches the full dataset $\mathcal{D}$. Post-filtering retrieves approximate nearest neighbors first and subsequently discards those failing the predicate. However, this strategy becomes *prohibitively expensive under low pass rates*, as the search must retrieve a massive volume of candidates to yield enough valid matches [27].

Joint-filtering has emerged as the mainstream paradigm, interleaving attribute evaluation directly with ANNS index traversal. It splits into two families, attribute-dedicated and attribute-agnostic, depending on when the attribute is reconciled with the vector space:

- Attribute-dedicated methods [48] bake the predicate into index topology at construction time, restructuring the graph into attribute-scoped sub-indexes. For example, FDiskANN [13] builds a subgraph for every label and stitches them into a unified index, iRangeGraph [52] targets numeric ranges by partitioning the attribute domain into a binary search tree of segments, building an independent subgraph per segment. However, this partitioning fails to establish reliable proximity edges across subgraphs, triggering routing failures under low pass rates.
- Attribute-agnostic methods [41, 42] instead maintain a predicate-blind index and inject the filter dynamically during search, enforcing explicit per-candidate checks over an otherwise ordinary graph. For example, ACORN [40] builds a standard HNSW graph and traverses it at query time. To maintain navigability across sparse valid candidates, it performs run-time multi-hop expansion, actively utilizing non-matching nodes as routing bridges to reach the next valid region. However, relying on these invalid bridge nodes inevitably wastes massive distance computations, severely degrading query efficiency under low pass rates.

In summary, the fundamental mismatch between continuous vectors and discrete attributes inherently disrupts the graph connectivity of existing FANNS methods under low pass rates. Lacking theoretical bounds that natively unify attribute filtering and vector distance estimation, these heuristic-driven methods suffer unpredictable performance degradation across varying pass rates. *This motivates us to design a novel FANNS method that adapts to diverse pass rates, underpinned by theoretical guarantees for exact attribute matching and accurate distance estimation.*

3 THE BASIC PART METHOD

In this section, we introduce the Phase-based Attribute Rotation Transform (PART), a novel FANNS method providing theoretical guarantees to overcome the unpredictable performance of existing approaches across varying pass rates. To clearly articulate the core mechanism, we first formalize the basic PART method assuming a *single* categorical attribute with an *exact-match* predicate. The extension of this foundational method to support multi-attribute filtering and complex predicates is detailed in §4.

3.1 Baking Attributes into Vectors

As summarized in Algorithm 1, consider a vector database $\mathcal{D} \subset \mathbb{R}^d$ where each data vector $\mathbf{o}$ is associated with an attribute value a from an attribute schema $\mathcal{A}$. We assume both data and query vectors

are *normalized* (i.e., $\|\mathbf{o}\|_2 = \|\mathbf{q}\|_2 = 1$). PART generates attribute-baked vectors that jointly capture vector similarity and attribute matching through a two-step pipeline: **(1)** Phase-Based Attribute Encoding: Translates the attribute value into a sequence of pseudo-random, unique phase angles. **(2)** Attribute-Baking Vector Rotation: Assembles these phase angles into an orthogonal matrix and applies it to rotate the vector $\mathbf{o}$, ultimately embedding the attribute value directly into the continuous vector space. We detail two steps below.

Phase-Based Attribute Encoding. Unlike existing FANNS methods that rely on decoupled data structures such as bitmaps or ID-lists to encode attributes, PART translates discrete attribute values into continuous phase angles. First, PART partitions the d vector dimensions into $B = d/2$ disjoint dimension pairs[1] (line 1 in Algorithm 1). Second, PART assigns each pair $j \in \{1, \ldots, B\}$ a salt s_j drawn independently from a common uniform distribution (line 2). The attribute value a is then mapped to a specific phase angle for the j-th pair using a pseudo-random hash function:

$$\psi_j(a) = 2\pi \cdot \mathrm{rand}(a, s_j) \in [0, 2\pi) \quad , \tag{3}$$

where $\mathrm{rand}(\cdot,\cdot)$ is a deterministic hash function that, over the random choice of salt s_j, outputs a value uniformly distributed in $[0, 1)$ (lines 6-8). Finally, by evaluating this hash function across all B pairs, PART concatenates the results to form a high-dimensional composite phase angle vector $\mathbf{p}$:

$$\mathbf{p}_a = \big(\psi_1(a), \ldots, \psi_B(a)\big)^\top \in [0, 2\pi)^B \quad . \tag{4}$$

This phase angle vector $\mathbf{p}_a$ serves as the unique representation for the attribute value. The set of random salts $\{s_j\}_{j=1}^B$ is sampled once offline during system initialization and shared globally across all data and query encodings. This mapping guarantees that identical attribute values evaluated under the same global salts deterministically map to the exact same high-dimensional angle vector, paving the way for the subsequent vector rotation.

Attribute-Baking Vector Rotation. PART translates the phase angle vector $\mathbf{p}$ into an orthogonal transformation matrix. Each phase angle $\psi_j(a)$ parameterizes a 2×2 standard rotation matrix for its respective j-th dimension pair. Collectively, these blocks form a d-dimensional, block-diagonal phase rotation matrix $\Phi(a) \in \mathbb{R}^{d\times d}$:

$$\Phi(a) = \bigoplus_{j=1}^{B} \mathbf{R}(\psi_j(a)), \quad \text{where} \quad \mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad . \tag{5}$$

Because every block is a standard two-dimensional rotation, the assembled matrix $\Phi(a)$ is strictly orthogonal. To bake the attribute value a into the data vector $\mathbf{o}$, PART simply applies this orthogonal rotation matrix to the vector: $\hat{\mathbf{o}} = \Phi(a)\mathbf{o}$ (line 12).

During the search phase, a normalized query vector $\mathbf{q}$ targeting attribute value t undergoes the exact same transformation: $\hat{\mathbf{q}} = \Phi(t)\mathbf{q}$. By applying these rotations, vectors equipped with identical attribute values ($a = t$) maintain their original spatial alignment, whereas vectors with differing attribute values ($a \neq t$) are forced into nearly orthogonal orientations. Since the relative rotation $\Phi(t)^\top\Phi(a)$ is governed by the per-pair angle differences $\psi_j(a) - \psi_j(t)$, which are random when $a \neq t$, the baked inner product (IP) concentrates near zero.

[1]Modern embedding models mostly produce even-dimensional vectors (e.g., Gemini [25] and Qwen3 [57]). PART thus assumes an even d. An odd d can be trivially padded with a zero without affecting the distance semantics.

Algorithm 1 Basic PART Method

Require: normalized vector database $\mathcal{D} \subset \mathbb{R}^d$, attribute schema $\mathcal{A}$
Ensure: attribute-baked vector database $\hat{\mathcal{D}}$
1: $B \leftarrow d/2$; ▷ Number of disjoint dimension pairs
2: Initialize global random salts $\{s_1, \ldots, s_B\}$;
3: $\hat{\mathcal{D}} \leftarrow \emptyset$;
4: **for each** vector $\mathbf{o} \in \mathcal{D}$, associated attribute value $a \in \mathcal{A}$ **do**
5: // Step 1: Phase-Based Encoding
6: **for** $j = 1$ **to** B **do**
7: $\psi_j(a) \leftarrow 2\pi \cdot \mathrm{rand}(a, s_j)$;
8: **end for**
9: $\mathbf{p} \leftarrow \big(\psi_1(a), \ldots, \psi_B(a)\big)^\top$; ▷ Composite phase angle vector
10: // Step 2: Attribute-Baking Rotation
11: $\Phi(a) \leftarrow \bigoplus_{j=1}^{B} \mathbf{R}(\psi_j(a))$; ▷ Assemble orthogonal matrix
12: $\hat{\mathbf{o}} \leftarrow \Phi(a)\mathbf{o}$; $\hat{\mathcal{D}} \leftarrow \hat{\mathcal{D}} \cup \hat{\mathbf{o}}$;
13: **end for**
14: **return** $\hat{\mathcal{D}}$

Example 1. *Consider a database with dimension $d = 4$. Let $\mathbf{o}_1 = \mathbf{o}_2 = (0.5, 0.5, 0.5, 0.5)^\top$ be two identical normalized vectors, where $\mathbf{o}_1$ is associated with attribute value $a_1 = 42$ and $\mathbf{o}_2$ with $a_2 = 99$. First, via Phase-Based Attribute Encoding, evaluating $a_1 = 42$ yields phase angles $\psi_1(42) = \pi/2$ and $\psi_2(42) = \pi$ for the two dimension pairs. These natively form the composite phase angle vector $\mathbf{p}_{a_1} = (\pi/2, \pi)^\top$. Through Attribute-Baking Vector Rotation, $\mathbf{p}_{a_1}$ parameterizes the orthogonal matrix $\Phi(a_1)$ as a direct sum of two 2×2 blocks:*

$$\Phi(a_1) = \mathbf{R}(\pi/2) \oplus \mathbf{R}(\pi) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad .$$

Conversely, the deterministic hash for $a_2 = 99$ yields a distinct angle vector $\mathbf{p}_{a_2} = (\pi, \pi/2)^\top$, assembling an independent rotation matrix:

$$\Phi(a_2) = \mathbf{R}(\pi) \oplus \mathbf{R}(\pi/2) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad .$$

The attribute values are then baked into data vectors: $\hat{\mathbf{o}}_1 = \Phi(a_1)\mathbf{o}_1 = (-0.5, 0.5, -0.5, -0.5)^\top$, and $\hat{\mathbf{o}}_2 = \Phi(a_2)\mathbf{o}_2 = (-0.5, -0.5, -0.5, 0.5)^\top$.

Given a normalized query $\mathbf{q} = (0.5, 0.5, 0.5, 0.5)^\top$ targeting attribute value $t = a_1 = 42$, it undergoes the exact same transformation $\hat{\mathbf{q}} = \Phi(a_1)\mathbf{q} = (-0.5, 0.5, -0.5, -0.5)^\top$. (1) For the matched vector $\mathbf{o}_1$ ($a_1 = t$), the relative rotation is $\Phi(a_1)^\top\Phi(a_1) = \mathbf{I}$. Their original IP is strictly preserved: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_1 \rangle = 1$. (2) For the mismatched vector $\mathbf{o}_2$ ($a_2 \neq t$), the relative rotation $\Phi(a_1)^\top\Phi(a_2)$ alters the spatial alignment. $\hat{\mathbf{q}}$ and $\hat{\mathbf{o}}_2$ are forced into orthogonality, penalizing their IP: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_2 \rangle = 0$, effectively pruning $\mathbf{o}_2$.

3.2 Effectiveness Theoretical Analysis

This section establishes the theoretical foundation explaining why PART is effective for FANNS. Specifically, we formalize its filtering mechanism through two guarantees: *attribute matching soundness* and *vector similarity completeness*. Respectively, they ensure the structural pruning of attribute-mismatched vectors, and the authentic distance evaluation for identically attributed vectors.

Theorem 1 (Attribute Matching Soundness). *Given a data vector $\mathbf{o}$ associated with a scalar attribute a, and a query vector $\mathbf{q}$*

targeting attribute t, assume data vectors and queries are normalized. If $a \neq t$, the baked IP has zero mean, $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$, with a variance bounded by $\mathrm{Var}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 1/d$. The tail probability that the mismatch IP exceeds a similarity threshold $\epsilon > 0$ decays exponentially with dimensionality d: $\mathbb{P}\left(\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle \geq \epsilon\right) \leq \exp\left(-\frac{d \cdot \epsilon^2}{4}\right)$.

Proof. Let $B = d/2$ be the number of 2D subspaces, and $\Delta_j = \psi_j(a) - \psi_j(t)$ be the phase difference on the j-th subspace. The baked IP decomposes into per-subspace contributions: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \sum_{j=1}^{B} X_j$, where $X_j = u_j \cos\Delta_j + v_j \sin\Delta_j$. Here, $u_j = \langle \mathbf{q}^{(j)}, \mathbf{o}^{(j)} \rangle$ and $v_j = \langle \mathbf{q}^{(j)}, J\mathbf{o}^{(j)} \rangle$, where J denotes the 90° rotation matrix. Since the hash is pairwise-independent, Δ_j is uniformly distributed in $[0, 2\pi)$. Thus, $\mathbb{E}[\cos\Delta_j] = \mathbb{E}[\sin\Delta_j] = 0$ and $\mathrm{Var}(\cos\Delta_j) = \mathrm{Var}(\sin\Delta_j) = 1/2$. With cross-terms vanishing due to independence, the expectation trivially evaluates to $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$. The variance of the sum is $\mathrm{Var} = \frac{1}{2}\sum_{j=1}^{B}(u_j^2 + v_j^2)$. By the 2D identity $\langle \mathbf{x}, \mathbf{y} \rangle^2 + \langle \mathbf{x}, J\mathbf{y} \rangle^2 = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$, we simplify the subspace variance to $u_j^2 + v_j^2 = \|\mathbf{q}^{(j)}\|^2 \|\mathbf{o}^{(j)}\|^2 \triangleq w_j$. When the per-subspace variance is balanced, each subspace uniformly carries $w_j = 1/B^2$. Consequently, the total variance evaluates exactly to $\frac{1}{2}\sum_{j=1}^{B} w_j = \frac{1}{2}\left(B \cdot \frac{1}{B^2}\right) = 1/d$. Finally, by standard trigonometric identities, $X_j = u_j \cos\Delta_j + v_j \sin\Delta_j$ is strictly bounded within $[-\sqrt{u_j^2 + v_j^2}, \sqrt{u_j^2 + v_j^2}]$. Since $w_j = u_j^2 + v_j^2$, we have $X_j \in [-\sqrt{w_j}, \sqrt{w_j}]$. Applying Hoeffding's Inequality yields the tail probability: $\mathbb{P}(\sum_{j=1}^{B} X_j \geq \epsilon) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{j=1}^{B}(2\sqrt{w_j})^2}\right)$. Since $\sum_{j=1}^{B}(2\sqrt{w_j})^2 = 4\sum_{j=1}^{B} w_j = 4B(1/B^2) = 4/B$, and substituting $B = d/2$, the bound simplifies to $\exp(-d \cdot \epsilon^2/4)$. □

The mismatch error bound can thus be concisely presented as $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = O(1/\sqrt{d})$ with high probability. This guarantees that false positives are structurally suppressed to a negligible noise floor.

Theorem 2 (Vector Similarity Completeness). *If a normalized data vector $\mathbf{o}$ matches the query attribute t (i.e., $a = t$), then their baked IP strictly equals the original IP: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$. Moreover, for any two matched vectors $\mathbf{o}_1, \mathbf{o}_2$ both with attribute t, $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_1 \rangle > \langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_2 \rangle \iff \langle \mathbf{q}, \mathbf{o}_1 \rangle > \langle \mathbf{q}, \mathbf{o}_2 \rangle$. Hence, the* ranking *of true matches under the baked space is identical to the unfiltered ranking.*

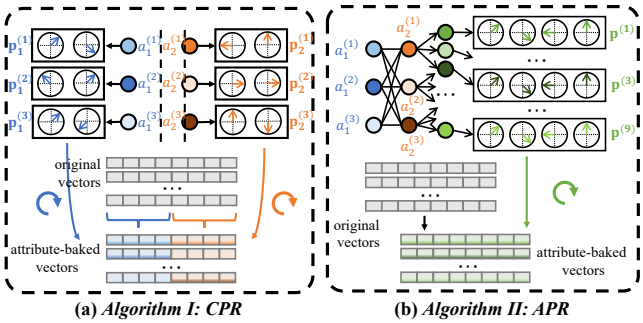
Proof. $a = t \Rightarrow \Phi(a) = \Phi(t)$. By definition, the baked IP is $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = (\Phi(t)\mathbf{q})^\top (\Phi(a)\mathbf{o}) = \mathbf{q}^\top \Phi(t)^\top \Phi(a)\mathbf{o}$. Since $\Phi(t)$ is an orthogonal rotation matrix, $\Phi(t)^\top \Phi(t) = \mathbf{I}$. Thus, $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \mathbf{q}^\top \mathbf{I}\mathbf{o} = \langle \mathbf{q}, \mathbf{o} \rangle$. For any two matched vectors, both $\hat{\mathbf{o}}_1$ and $\hat{\mathbf{o}}_2$ are obtained by applying the same orthogonal transform $\Phi(t)$. Since orthogonal transforms preserve IPs and distances, the order of $\langle \hat{\mathbf{q}}, \cdot \rangle$ over the matched set exactly equals the order of $\langle \mathbf{q}, \cdot \rangle$. □

4 THE EXTENSION OF PART

In this section, we extend PART beyond single-attribute equality filtering to support complex predicates, including multi-attribute AND, continuous numeric ranges, and arbitrary mixtures of both.

4.1 Multi-Attribute AND Extension

We design two baking mechanisms to support AND predicates within PART. We first describe an intuitive block-partitioning solution



(a) *Algorithm I: CPR* (b) *Algorithm II: APR*

Figure 1: Illustration of CPR & APR for multi-attribute AND.

(*Algorithm I*), and then present a global-superposition mechanism (*Algorithm II*) that provides rigorous theoretical guarantees.

Algorithm I: Composite Phase Rotation (CPR). CPR partitions the vector space into multiple sub-blocks, applying single-attribute rotations to each sub-block. This approach decomposes a multi-attribute AND into a series of localized, single-attribute filters. As shown in Figure 1(a), for a data vector tagged with an M-attribute tuple $\mathbf{a} = (a_1, \ldots, a_M)$, CPR evenly partitions the $d/2$ dimension-pairs into M disjoint blocks $\{\mathcal{B}_1, \ldots, \mathcal{B}_M\}$. CPR independently maps each attribute value a_m to its designated block $\mathcal{B}_m$ to derive a localized angle vector $\mathbf{p}_{a_m}$, where $\psi_j(a_m)$ extends Eq. (3) to the m-th attribute value. These sub-rotations are then assembled into a unified block-diagonal matrix: $\Phi(\mathbf{a}) = \bigoplus_{m=1}^{M}\left(\bigoplus_{j \in \mathcal{B}_m} \mathbf{R}(\psi_j(a_m))\right)$. By applying the exact same partitioning to the query vector targeting $\mathbf{t} = (t_1, \ldots, t_M)$, a data vector fully aligns with the query if and only if all attributes match ($a_m = t_m$ for all m), organically embedding AND into the vector space.

Example 2. *Consider a $d = 8$ space where we execute a multi-attribute* AND *query targeting $M = 2$ attributes. CPR partitions the 4 dimension-pairs into $\mathcal{B}_1$ (pairs 1-2) for the first attribute and $\mathcal{B}_2$ (pairs 3-4) for the second. For a data vector $\mathbf{o}$ tagged with attribute tuple $\mathbf{a} = (42, 99)$ (i.e., $a_1 = 42, a_2 = 99$), its localized phase signatures (e.g., $\mathbf{p}_{a_1} = (\pi/2, \pi)$ and $\mathbf{p}_{a_2} = (\pi, \pi/2)$) directly assemble the unified rotation matrix: $\Phi(\mathbf{a}) = \mathbf{R}(\pi/2) \oplus \mathbf{R}(\pi) \oplus \mathbf{R}(\pi) \oplus \mathbf{R}(\pi/2)$. The vector is then baked: $\hat{\mathbf{o}} = \Phi(\mathbf{a})\mathbf{o}$. Given a query vector $\mathbf{q}$ where $\langle \mathbf{q}, \mathbf{o} \rangle = 1$, when* AND *is satisfied, querying $\mathbf{t} = (42, 99)$ applies identical rotations across all blocks, fully preserving the score $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = 1$. Conversely, when* AND *fails, querying $\mathbf{t}' = (42, 77)$ aligns only on $\mathcal{B}_1$.*

However, CPR compromises soundness (Theorem 1) due to dimensionality dilution. While Theorem 1 suppresses false positives via the exponential bound $\exp(-d \cdot \epsilon^2/4)$ over the full dimensionality d, CPR partitions the space and reduces the effective dimensionality of each block to d/M. This structural division creates a vulnerability for near-miss vectors (e.g., matching $M - 1$ attributes but failing on exactly one). In such cases, orthogonal penalization occurs *only* within the single mismatched block, degrading the mismatch error bound from $O(1/\sqrt{d})$ to $O(\sqrt{M/d})$. Meanwhile, the mismatched block, suffering from reduced dimensionality, fails to generate sufficient orthogonal penalty to push the overall score below the similarity threshold. This leads to an increase in false positives for AND queries with a large number of attributes.

Algorithm II: Additive Phase Rotation (APR). APR superimposes the phase angles of all M attributes over *every* 2D subspace.

As shown in Figure 1(b), for a data vector labeled with an M-attribute tuple $\mathbf{a} = (a_1, \ldots, a_M)$, APR aggregates the per-attribute phases, each modulated by an independent salt $s_m(j)$, to compute a phase angle for each dimension pair j. Here $\psi_{m,j}(a_m) = 2\pi \cdot \mathrm{rand}(a_m, s_m(j))$ extends Equation (3) to the m-th attribute, and the salts $\{s_m(j)\}$ are drawn i.i.d. across both m and j, independent of the base salts $\{s_j\}$. The rotation matrix is then constructed:

$$\Phi(\mathbf{a}) = \bigoplus_{j=1}^{d/2} \mathbf{R}\Big(\sum_{m=1}^{M} s_m(j)\psi_{m,j}(a_m) \Big) \quad . \tag{6}$$

By applying the exact same superposition to the query targeting $\mathbf{t} = (t_1, \ldots, t_M)$, let $\delta_{m,j} = \psi_{m,j}(a_m) - \psi_{m,j}(t_m)$ denote the phase difference for the m-th attribute on pair j. The total relative phase shift between the baked data vector and the query strictly evaluates to $\Delta_j = \sum_{m=1}^{M} s_m(j)\delta_{m,j}$.

Example 3. *Revisit the $d = 8$ space with an $M = 2$ attribute query. APR superimposes both attributes across* all *4 pairs. For a data vector* $\mathbf{o}$ *tagged with attribute tuple* $\mathbf{a} = (42, 99)$ *(i.e., $a_1 = 42, a_2 = 99$), the angle on any pair $j \in \{1, 2, 3, 4\}$ is a salted sum of both phases: $s_1(j)\psi_{1,j}(a_1) + s_2(j)\psi_{2,j}(a_2)$. Given a query* $\mathbf{q}$ *where $\langle \mathbf{q}, \mathbf{o} \rangle = 1$, when* AND *is satisfied, querying* $\mathbf{t} = (42, 99)$ *yields zero phase differences ($\delta_{1,j} = \delta_{2,j} = 0$) globally, fully preserving the score $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = 1$. Conversely, when* AND *fails, querying* $\mathbf{t}' = (42, 77)$ *matches the first attribute but generates a random non-zero difference $\delta_{2,j}$ for the second. Because APR superimposes phases, the total shift $\Delta_j = s_2(j)\delta_{2,j} \neq 0$ affects* every *pair. Moreover, since $\delta_{2,j}$ varies randomly across j, these shifts are independent across pairs, thereby scrambling the entire vector space simultaneously upon any single mismatch.*

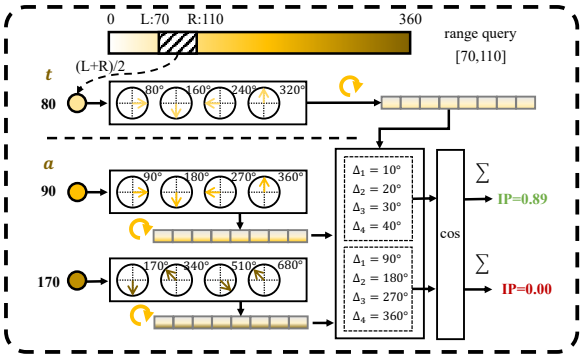
By superimposing the rotations of all M attributes across *every* dimension-pair j, APR eliminates the dimensionality dilution vulnerability inherent in CPR. Concretely, when $\mathbf{a} = \mathbf{t}$ all per-attribute differences $\delta_{m,j}$ vanish, so $\Delta_j = 0$ for every pair and the IP is preserved exactly; when some attribute mismatches, the resulting Δ_j remains a random phase over all $B = d/2$ pairs, so the concentration argument of Theorem 1 applies over the full dimensionality d rather than d/M. We therefore adopt APR as the foundational mechanism for subsequent extensions.

4.2 Continuous Numeric Ranges Extension

Unlike exact-match predicates that rely on random hashing to penalize arbitrary mismatches, range filtering needs to preserve numerical locality to construct a valid bounding window. To this end, we next introduce Scaled Phase Rotation (SPR).

Algorithm III: Scaled Phase Rotation (SPR). Unlike exact-match predicates, which rely on the random phase $\psi_j(a)$ to penalize arbitrary mismatches, range filtering must preserve numerical locality so that nearby values remain similar. SPR anchors the baseline rotation at the range center $t = (l + u)/2$ and translates deviations from t into proportional angular shifts: values *inside* $[l, u]$ undergo bounded rotations that preserve similarity, whereas values *outside* accumulate destructive phase offsets. Specifically, for a numeric attribute $a \in [0, L]$ (L its value range), the range phase on the j-th dimension-pair ($j \in \{1, \ldots, B\}$, $B = d/2$) is

$$\phi_j(a) = \pi \frac{j}{B} \frac{a}{L} \quad . \tag{7}$$



Algorithm III: SPR

Figure 2: Illustration of SPR for range attribute.

The relative shift against t is $\Delta_j = \phi_j(a) - \phi_j(t) = \pi \frac{j}{B} \frac{a-t}{L}$. For $a \in [l, u]$, $|\Delta_j| \leq \pi \frac{u-l}{2L}$, which is small when $u - l \ll L$, so $\cos \Delta_j \approx 1$ and the inner product is largely preserved; as a moves away from t, Δ_j grows linearly with $|a - t|$ and with j/B, so higher-index pairs yield larger shifts and out-of-window vectors are progressively down-ranked. When $u - l$ is not small relative to L, the query partitions $[l, u]$ into $m = \lceil (u - l)/w \rceil$ sub-queries of width $w \ll L$ and merges their candidate sets.

Example 4. *Consider a $d = 8$ space ($B = 4$ pairs) with value range $[0, 100]$, a range query $[40, 60]$ centered at $t = 50$, and three data vectors $\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_3$ with $\langle \mathbf{q}, \mathbf{o}_i \rangle = 1$. Assuming even energy across the 4 pairs, the baked score is $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_i \rangle = \frac{1}{4} \sum_{j=1}^{4} \cos \Delta_j$. For the center $\mathbf{o}_1$ ($a_1 = 50$), $\Delta_j = 0$ for all j, so the score stays 1. For an in-window boundary vector $\mathbf{o}_2$ ($a_2 = 60$), the offset 10 gives $\Delta_j = \pi \frac{j}{40}$ and a score ≈ 0.98, slightly attenuated but safely retrieved. For an out-of-window vector $\mathbf{o}_3$ ($a_3 = 90$), the offset 40 gives $\Delta_j = \pi \frac{j}{10}$ and $\Delta_4 = 0.4\pi$, lowering the score to ≈ 0.66 — clearly below the in-window case, though not collapsed to zero.*

SPR's primary theoretical advantage is *bounded IP preservation*, providing provable in-window fidelity over unpredictable heuristic degradation. We formalize this via two bounds. **(1)** For a single query (m=1), the maximum phase shift is bounded by $\pi w/(2L)$, so the in-window IP deviation is $O((w/L)^2)$. For narrow ranges (e.g., $w/L \leq 3\%$), the resulting attenuation is below 0.2%, generating high-fidelity candidates for a lightweight rerank. **(2)** For wider ranges, *sub-query partitioning* tightens the bound: executing m equal-width sub-queries compresses the maximum phase shift to $\pi w/(2mL)$, so the in-window IP converges to the exact semantic IP as m grows. By scaling m with the window width to bound the residual error, SPR preserves the scores of in-window members, enabling the reranker to recover the exact Top-k given a sufficiently high-recall ANN index. Formal proofs are in Appendix A.

4.3 Hybrid Predicates Extension

Real-world applications frequently demand the simultaneous evaluation of discrete categories and continuous numerical ranges. To fuse these heterogeneous filtering within a single space, we introduce Hybrid Additive Phase Rotation (HAPR).

Algorithm IV: Hybrid Additive Phase Rotation (HAPR). Built upon the additive phase superposition paradigm, HAPR seamlessly integrates APR and SPR into every 2D subspace. Specifically,

consider a hybrid query comprising A discrete attributes $\mathbf{t}^{\text{disc}} = (t_1^{\text{disc}}, \ldots, t_A^{\text{disc}})$ and C range attributes $\{[l_1, u_1], \ldots, [l_C, u_C]\}$. For each numeric range $[l_k, u_k]$, we target its center $t_k^{\text{num}} = (l_k + u_k)/2$. HAPR computes a composite phase angle for each dimension-pair $j \in \{1, \ldots, B\}$ by aggregating the salted phase shifts of the discrete attributes and the dimension-dependent angular shifts of the numeric attributes. The unified rotation matrix is constructed as:

$$\Phi(\mathbf{t}) = \bigoplus_{j=1}^{B} \mathbf{R}\left(\sum_{m=1}^{A} s_m(j)\psi_{m,j}^{\text{disc}}(t_m^{\text{disc}}) + \sum_{k=1}^{C} \psi_{k,j}^{\text{num}}(t_k^{\text{num}})\right) \quad . \tag{8}$$

where $\psi_{k,j}^{\text{num}}(a) = \pi \frac{j}{B} \frac{a}{L_k}$ denotes the scaled phase for the k-th numeric attribute defined over $[0, L_k]$. Let $\delta_{m,j}^{\text{disc}}$ and $\delta_{k,j}^{\text{num}}$ denote the relative phase differences for the m-th discrete and k-th numeric attributes, respectively. The total relative phase shift between a baked data vector and the hybrid query becomes:

$$\Delta_j = \sum_{m=1}^{A} s_m(j)\delta_{m,j}^{\text{disc}} + \sum_{k=1}^{C} \delta_{k,j}^{\text{num}} \quad . \tag{9}$$

Example 5. *Revisit the $d = 8$ space ($B = 4$ dimension-pairs). We execute a hybrid query comprising one exact-match attribute and one numeric attribute over the domain $[0, 100]$. Given a data vector $\mathbf{o}$ tagged with $a^{disc} = 42$ and $a^{num} = 55$, alongside a query vector $\mathbf{q}$ where $\langle \mathbf{q}, \mathbf{o} \rangle = 1$, HAPR exhibits four distinct filtering behaviors: (1) Full Match Preservation. The query targets 42 and the range $[50, 60]$ (center $t^{num} = 55$). Both phase differences are globally zero. The relative rotation is the identity matrix $\mathbf{I}$, preserving the score $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = 1$. (2) Exact-Match Violation. The query targets 77 and $[50, 60]$. While the numeric center matches, the exact-match attribute diverges. The salted offset introduces random phase shifts across all 4 pairs, fully scrambling the space and collapsing the score toward the $O(1/\sqrt{d}) \approx 0$ error bound. (3) Numeric Deviation. The query targets 42 and $[10, 20]$ (center $t^{num} = 15$). The exact-match attribute aligns, but the numeric value deviates by 40. The dimension-dependent angular shift grows linearly, reaching a maximum of $\Delta_4 = 0.4\pi$. The overall score gracefully attenuates to $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle \approx 0.43$, effectively downranking the vector. (4) Combined Mismatch. The query targets 77 and $[10, 20]$. Both mismatches contribute to the total phase shift. These perturbations superimpose, aggressively pruning the candidate and suppressing the score far below either single-mismatch scenario.*

4.4 Summary and Discussion

We extend the base PART method with advanced mechanisms like APR, SPR, and HAPR. As summarized in Table 3, PART systematically broadens its capability to support complex hybrid filtering conditions (e.g., multi-attribute `AND` and continuous numeric ranges). This unified geometric framework ensures rigorous multidimensional filtering without sacrificing fundamental search accuracy.

Remark1. PART relies on an offline baking process that embeds attributes into the vector. This inherently dictates that the schema of filtering attributes must be pre-defined; PART cannot support on-the-fly, ad-hoc filtering for arbitrary attributes that were not explicitly baked in advance. In scenarios with highly dynamic attribute schemas, PART is best utilized as an aggressive primary filter—baking the most critical or frequently queried attributes (e.g., category, tenant ID, or timestamp) to structurally prune the majority of the search space. Furthermore, PART can be seamlessly incorporated into low-overhead index structures, such as quantization, to facilitate FANNS rapid pruning, which we detail in §5.1.

Table 3: Predicates supported by PART.

Predicate	Formal Definition	Error Guarantee
EQ	$a = t$	$O(1/\sqrt{d})$
AND	$a_1 = t_1 \wedge \cdots \wedge a_A = t_A$	$O(1/\sqrt{d})$
Range	$l \le a \le u$	$\le \pi s w/(2L)$
Hybrid	$a^{\text{disc}}{=}t^{\text{disc}} \wedge l \le a^{\text{num}} \le u$	$\le \pi s w/(2L)$
OR / IN	DNF expressions over attributes	Inherited per branch

Remark2. PART does not natively support disjunctive predicates such as `OR` and `IN`. To address this, one viable solution is to adopt a system-level *query-side splitting* strategy. Since any complex Boolean filtering condition can be normalized into Disjunctive Normal Form (DNF), we decompose the original query into multiple purely conjunctive subqueries. Each subquery is encoded independently, leveraging PART's native support for conjunctions. The final result is obtained by taking the union of matches returned by all subqueries, followed by deduplication to eliminate overlapping candidates across different branches. As this system-level routing strategy is beyond the primary scope of this paper, we leave its systematic exploration and optimization to future work.

5 IMPLEMENTATIONS OF PART

5.1 RaBitQ Integration

To achieve high efficiency in large-scale ANNS, VQ is indispensable for reducing memory footprint and accelerating distance computations. PART seamlessly integrates with state-of-the-art VQ algorithm, RaBitQ, to support both accurate distance estimation and exact attribute matching with theoretical guarantees.

We employ RaBitQ as the underlying quantization framework due to its profound compatibility with PART. First, both architectures rely fundamentally on orthogonal transformations. RaBitQ utilizes a random orthogonal matrix $\mathbf{P}$ to equalize dimensional variances, satisfying the key requirement of PART for balanced pairwise variance across subspaces. This deep compatibility inherently ensures that IPs of attribute mismatched vectors converge to the $O(1/\sqrt{d})$ noise floor. Second, their transformation operations are fusible. The attribute rotation $\mathbf{\Phi}(a)$ of PART and the projection matrix $\mathbf{P}$ of RaBitQ can be precomputed offline into a unified operator $\mathbf{P}^\top \mathbf{\Phi}(a)$. Consequently, PART achieves exact attribute filtering and establishes tight distance error bounds with zero additional encoding overhead relative to RaBitQ. Third, because the phase modulation mechanism of PART strictly preserves vector norms, the integrated method retains the inherent capability of RaBitQ to accurately reconstruct L_2 distances directly from estimated IPs. We next detail the three core steps of this integrated pipeline.

Baked Binary Quantization. The offline index construction of RaBitQ-PART proceeds in three sequential operations: vector preprocessing, fused vector rotation, and binary quantization. **(1)** Vector Preprocessing: Given a raw data vector $\mathbf{o}_r$, we first apply ℓ_2-normalization to obtain the unit vector $\mathbf{o} = \mathbf{o}_r / \|\mathbf{o}_r\|_2$. **(2)** Fused Vector Rotation: For the normalized vector $\mathbf{o}$ and its associated attribute value a, PART first maps the attribute to its corresponding rotation matrix $\mathbf{\Phi}(a)$. To fulfill the dimensional variance balancing

required by RaBitQ, PART incorporates a shared random orthogonal matrix $\mathbf{P}$. They are algebraically pre-multiplied into a single unified operator $\mathbf{P}^\top\Phi(a)$. **(3)** Binary Quantization: The fundamental objective of binary quantization is to assign the vector to its nearest codeword in the shared base codebook $C = \{\pm 1/\sqrt{d}\}^d$. The optimal codeword reduces directly to taking the element wise sign of the projected vector: $\bar{\mathbf{x}}_i = \text{sign}\big(\mathbf{P}^\top\Phi(a)\mathbf{o}\big) \in \{\pm 1\}^d$.

Distance Estimation and Hardware Acceleration. During the online retrieval phase, PART evaluates the similarity of the baked vectors. Because attribute mismatches inherently collapse to the noise floor under the rotated representation, RaBitQ-PART explicitly focuses on rapidly evaluating the continuous distance for all candidates without requiring explicit conditional filtering branches. (1) Distance Estimation: The original squared Euclidean distance can be reconstructed using the standard geometric expansion. By substituting the unbiased IP estimate derived from the baked vectors into this expansion alongside the original norms, we derive the final distance estimator: $||\mathbf{o}_i - \mathbf{q}||_2^2 \approx ||\mathbf{o}_i||_2^2 + ||\mathbf{q}||_2^2 - 2\widehat{\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_i \rangle}$. Because the IP estimation error is strictly bounded to $O(1/\sqrt{d})$, the recovered continuous distances faithfully preserve the ranking of valid matches within the theoretical error bounds. (2) Hardware Implementation: PART seamlessly inherits the SIMD accelerated bitwise operations of standard RaBitQ. The unbiased estimate of the baked inner product is rapidly computed via a hardware friendly formulation: $\widehat{\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_i \rangle} = \frac{\langle \bar{\mathbf{x}}_i, \mathbf{P}^\top \hat{\mathbf{q}} \rangle}{c_i}$. This branch free bitwise execution unifies soft range filtering and continuous distance approximation into a single hardware optimized scan, maximizing execution throughput.

5.2 Index Integration

Because PART maps discrete attribute logic into continuous vector geometry, it operates as a transparent coordinate transformation. This allows PART to seamlessly integrate with standard ANNS indexes, including both IVF and Graph-based indexes. We design two deployment paradigms to support different system requirements: building dedicated indexes on the baked space, and executing predicate-agnostic routing on the raw vector space.

Paradigm I: Indexing on the Baked Space. For workloads dominated by filtered queries, the most straightforward approach is to replace the original data vectors with their PART-baked counterparts ($\hat{\mathbf{o}}$) during index construction. Because PART preserves distances for identically attributed vectors while orthogonally repelling mismatched vectors to the noise floor, vectors sharing the same attributes naturally cluster together in the high-dimensional baked space. Standard graph indexes like HNSW and NSG can be constructed directly on this baked space without any modification to their routing logic. Furthermore, for quantization-supported frameworks such as IVF-RaBitQ and SymphonyQG, the system can directly ingest the RaBitQ-PART binary codes. During retrieval, the index leverages the hardware-friendly bitwise estimator to execute joint distance evaluation and attribute filtering in a single branch-free scan, aggressively accelerating search throughput.

Paradigm II: Indexing on the Raw Space. To support mixed workloads of both pure ANNS and FANNS, PART supports execution over graph or IVF constructed purely on the raw vectors. When a filtered query is issued, the graph traversal (or cluster probing) relies on the original vector neighborhood structure to navigate toward the target region. During the node evaluation phase, the system intercepts the standard distance computation and injects the RaBitQ-PART two-stage estimator:

- **Fast Bitwise Pruning:** Using the pre-computed binary codes, PART evaluates the baked IP on the fly. Attribute-mismatched candidates organically collapse to $O(1/\sqrt{d})$ and are pruned from the queue, while valid matches yield accurate estimated distances.
- **Post-Retrieval Reranking:** The top-k candidates surviving the bitwise filter undergo a lightweight reranking step using the raw vectors and exact attributes. This step resolves the soft attenuation of range predicates and calibrates the final ranking.

The primary advantage of this raw-space integration is its zero-duplication memory efficiency. A single raw graph or IVF index seamlessly serves both standard similarity search and complex filtered queries, with PART acting purely as a high-speed, hardware-accelerated evaluator during the candidate assessment phase.

Table 4: Vector dataset statistics.

Vector Dataset	Dimension	Data Size	Query Size	Type
SIFT	128	1,000,000	10,000	Image
MS MARCO	1024	1,000,000	1,000	Text
OpenAI	1536	999,000	1,000	Text
ArXiv	4096	1,000,000	10,000	Text

Table 5: Attribute dataset statistics.

Workload	Predicate	Selectivity	Type
A1_C2	$a_1 = t_1$	50%	Categorical
A1_C10	$a_1 = t_1$	10%	Categorical
A2_C10	$(a_1 = t_1) \wedge (a_2 = t_2)$	1%	Categorical
A3_C10	$\bigwedge_{j=1}^{3}(a_j = t_j)$	0.1%	Categorical
ArXiv_Attr	$\bigwedge_{j=1}^{12}(a_j = t_j)$	$\approx$ 30%	Categorical
A1_W50	$\ell_1 \le r_1 \le u_1$	50%	Numeric
A1_W10	$\ell_1 \le r_1 \le u_1$	10%	Numeric
A2_W10	$\bigwedge_{j=1}^{2}(\ell_j \le r_j \le u_j)$	1%	Numeric
A3_W10	$\bigwedge_{j=1}^{3}(\ell_j \le r_j \le u_j)$	0.1%	Numeric
A2_H10	$(a = t) \wedge (\ell \le r \le u)$	1%	Hybrid

6 EXPERIMENTAL EVALUATION

We evaluate the following aspects of PART: (1) search performance of equality, continuous numeric range, and hybrid filters (**§6.2**), (2) index cost (**§6.3**), (3) ablation study (**§6.4**), (4) parameter sensitivity (**§6.5**), and (5) empirical validation of theoretical bounds (**§6.6**). Kindly refer to Appendix for additional evaluations.

6.1 Experimental Settings

Datasets. We evaluate PART on four public ANNS datasets: SIFT[2], MS MARCO[3], OpenAI[4], and ArXiv[5]. These datasets span a wide range of dimensionality from 128 to 4096. Table 4 summarizes their dimensionality, database size, query set size, and data modalities.

For every vector dataset, we construct equivalent attribute workloads including equality, continuous numeric ranges, and mixed

[2] http://corpus-texmex.irisa.fr/
[3] https://huggingface.co/datasets/CohereLabs/msmarco-v2.1-embed-english-v3
[4] https://huggingface.co/datasets/Qdrant/dbpedia-entities-openai3-text-embedding-3-large-1536-1M
[5] https://hf.co/datasets/SPCL/arxiv-for-fanns-large

filters, as summarized in Table 5. The equality workload assigns each vector A categorical attributes sampled independently and uniformly from a domain of cardinality C, yielding an expected selectivity of C^{-A}. We instantiate four equality workloads, specifically A1_C2, A1_C10, A2_C10, and A3_C10, covering selectivities from 50% down to 0.1%. These attribute streams are generated independently of the vector datasets. This setting prevents any method from exploiting alignments between similarity neighborhoods and attribute values. We similarly construct continuous numeric range workloads using a numeric attribute with a domain cardinality of 100, specifying contiguous intervals W to target the exact same selectivity distribution. The mixed workload equips every object with both one categorical attribute and one numeric attribute, combining an equality condition with a numeric bound via a strict logical conjunction to target a 1% selectivity.

Finally, ArXiv [20] serves as the dataset shipping with a real attribute, where each object carries a native ArXiv sub category label. We extract this categorical attribute set and apply it to the remaining three vector datasets as an additional equality workload, yielding an average selectivity of approximately 30%. Unlike synthetic generation, the selectivity is governed entirely by a *highly skewed* category distribution. This non uniform long tail distribution presents a distinctly challenging evaluation scenario, complementing our synthetic workloads by rigorously testing algorithm robustness against natural data skew.

Algorithms. We compare PART against prominent IVF-based and graph-based FANNS methods as follows: **(1) IVF-RaBitQ PostFilter** and **(2) IVF-RaBitQ InFilter** serve as baselines derived from IVF-RaBitQ [12]. The former evaluates constraints after vector similarity search, whereas the latter evaluates them on the fly for each candidate during the search process. **(3) FAISS-IVF** [23] executes predicate agnostic IVF scans using an external ID selector. **(4) RangePQ** [55] augments a shared PQ index with attribute ordered trees, specializing exclusively in dynamic continuous numeric ranges. **(5) SymphonyQG InFilter** serves as a controlled baseline derived from SymphonyQG [15], evaluating constraints on the fly for each candidate during its quantization aware graph traversal. **(6) ACORN** [40] is a predicate agnostic HNSW variant that dynamically executes predicate subgraph traversals to support equality, range, and Boolean filters. **(7) NHQ** [48] builds a composite proximity graph fusing vector and attribute distances, specializing strictly in conjunctive equality over fixed categorical attributes. **(8) FDiskANN** [13] incorporates discrete labels into Vamana construction for label aware pruning, natively supporting only single categorical equality. Our **(9) IVF-RaBitQ PART** and **(10) SymphonyQG PART** natively bake attribute constraints into the quantization space. By integrating the phase rotation framework into both inverted files and the quantization aware graph search of SymphonyQG [15], these proposed implementations seamlessly support equality, multi attribute AND, and continuous numeric ranges.

Metrics. We report the Recall@100 defined in Equation 1 alongside the scan throughput measured in QPS.

Parameter Setting. To ensure a fair comparison, every method is evaluated using identical generated queries, attributes, and ground truth. We detail the index construction and parameter configurations below. (1) For IVF-based methods, both the RaBitQ variant and the FAISS-IVFPQ baseline share a coarse quantizer with nlist = 4096. All RaBitQ variants encode each vector with a 1 bit code and a query refinement $B_{\text{query}} = 4$. In contrast, FAISS-IVFPQ uses $M = D/4$ for all datasets, allocating 8 bits per subspace. To measure the performance tradeoff between recall and QPS, we sweep the search budget by varying nprobe from 16 to 1500. (2) For graph based methods, the RNG-based methods, including FDiskANN, iRangeGraph, and SymphonyQG, all maintain a maximum degree $R = 32$. The NHQ baseline is built on KGraph with a maximum degree $R = 100$. To establish the same performance tradeoff, we sweep the search budget by scaling the search list L_{search} up to 1000.

Experimental Setup. We use C++ implementations of all methods for the index construction and search stages. All evaluations are carried out on a server equipped with dual AMD EPYC 9534 processors, which provide 128 physical cores, 256 GB of memory, and full support for the AVX-512 instruction set. Although index construction exploits the multi-core architecture, search runs with a single thread per query. We report the averaged results from three independent trials using the optimal configuration.

6.2 Overall Performance Evaluation

We present a comprehensive evaluation comparing the search performance of our proposed PART against state-of-the-art baselines, evaluated through the trade-off between QPS and Recall@100. In the presented trade-off grids, curves positioned closer to the top-right corner indicate superior performance. Figure 3 illustrates the experimental results across four datasets: SIFT, MS MARCO, OpenAI, and ArXiv. For each dataset, we evaluate five distinct workloads with varying selectivities ranging from 50% down to 0.1%. Notably, the ArXiv-Attr workload with 30% selectivity features a highly skewed attribute distribution.

Superiority of PART across Baselines. Across all tested datasets and index architectures, the methods equipped with our proposed framework, SymphonyQG PART and IVF-RaBitQ PART, consistently dominate the upper-right corner of the trade-off grids. For instance, on the high-dimensional OpenAI dataset, SymphonyQG PART significantly outperforms state-of-the-art graph-based methods like ACORN and FDiskANN, achieving up to 2× to 5× higher QPS at a strict 90% Recall@100 target. Similarly, IVF-RaBitQ PART drastically improves over its InFilter and PostFilter counterparts (indicated by dashed lines in Figure 3).

Performance under High to Moderate Selectivities. Under relatively unrestrictive conditions (50% and 10% selectivities), standard filtering approaches can typically locate enough valid candidates. Consequently, the performance of FANNS algorithms primarily depends on the navigation capability of the underlying index, allowing most baselines to rapidly achieve over 90% recall. Nevertheless, because PART seamlessly converts predicate verification into geometric distance calculation, both SymphonyQG PART and IVF-RaBitQ PART attain superior search efficiency compared to traditional methods.

Robustness under Extreme Low Selectivities. However, under extreme low selectivities, the degree of integration between

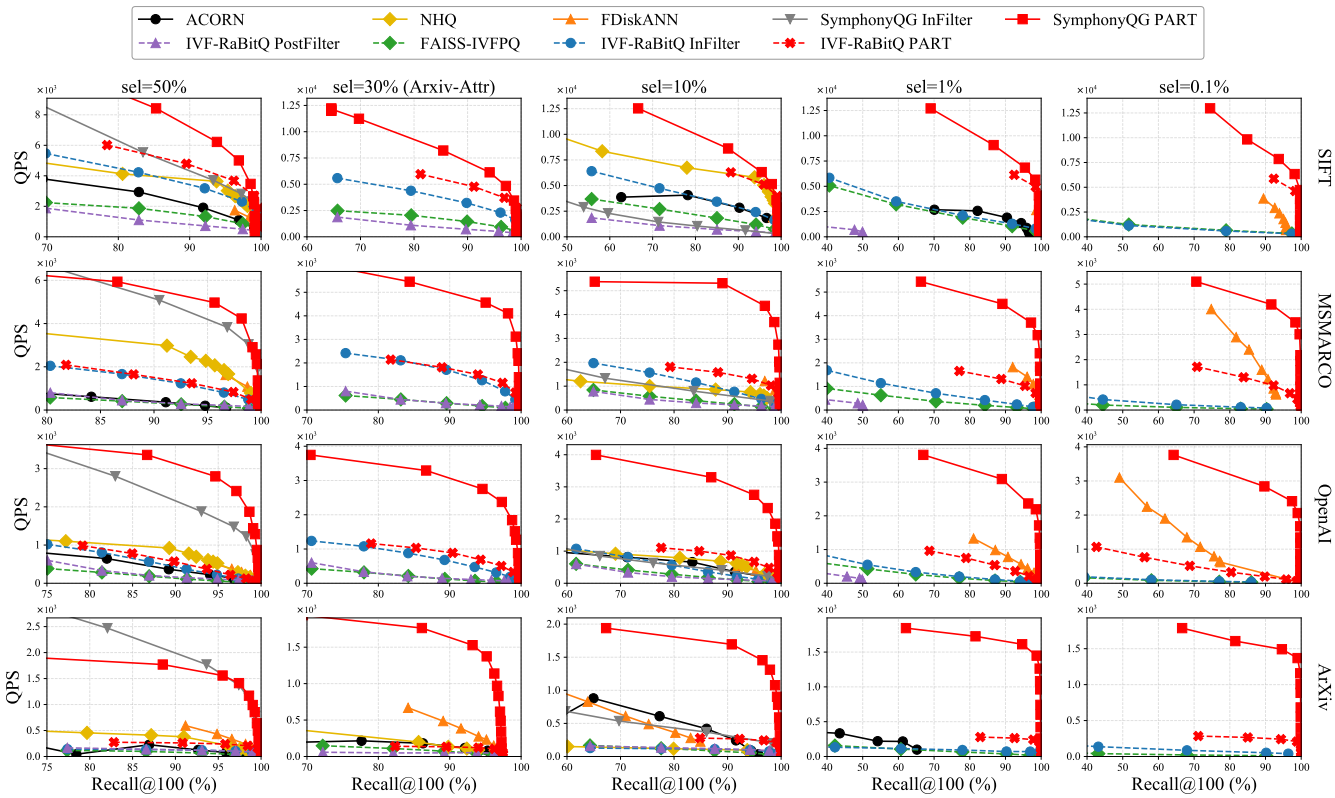


Figure 3: Filtered Recall@100–QPS trade-offs for the IVF-based methods on three vector datasets (rows) and four equality workloads with target selectivities of 50%, 10%, 1%, and 0.1% (columns). Each point is one evaluated search-effort configuration; upper right is better. Axes are scaled independently for readability, so comparisons should be made within a panel. A curve absent from a panel has no measured point within the displayed recall range and should not be interpreted as zero throughput.

attributes and vectors becomes critical. Conventional indices like ACORN and even dedicated indices like NHQ severely struggle to meet the recall targets. In contrast, both SymphonyQG PART and IVF-RaBitQ PART successfully achieve over 99% recall. Notably, SymphonyQG PART significantly outperforms FDiskANN, while IVF-RaBitQ PART achieves search performance comparable to FDiskANN. This success stems from the attribute constraints are embedded as orthogonal rotations/dimensions, vectors failing the predicate are geometrically pushed away in the high-dimensional space. Consequently, the underlying ANN routing mechanisms (both graph-based and IVF-based) naturally bypass invalid clusters without needing explicit conditional checks. The rightmost columns of Figure 3 demonstrate that even at 0.1% selectivity, SymphonyQG PART and IVF-RaBitQ PART maintain remarkably high QPS and stable recall curves, successfully insulating the vector search from the detrimental effects of restrictive combinatorial attribute constraints.

Search Performance for Complex Predicates.

6.3 Index Cost

For generic methods (e.g., ACORN) the values report the exact index size, for attribute-dedicated methods (e.g., NHQ), the values are averaged over different attribute workloads.

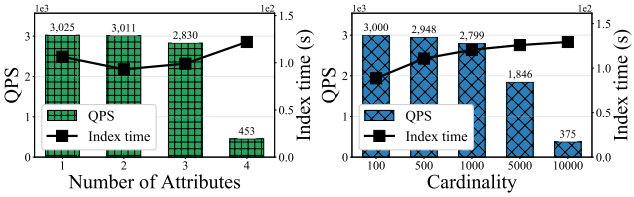


Figure 4: Parameter sensitivity of PART.

6.4 Ablation Study

6.5 Parameter Sensitivity

To evaluate the robustness of PART under different workload conditions, we conduct a parameter sensitivity analysis focusing on two key factors: the number of attributes and the attribute cardinality. All evaluations in this section are conducted on the OpenAI dataset using the SymphonyQG PART. The results, detailing both QPS (measured at a target Recall@100 of 90%) and index time, are presented in Figure 4. In our experiments, we isolate the effect of each parameter by fixing the other: when evaluating the number of queried attributes, the attribute cardinality is fixed at 10; conversely, when evaluating cardinality, the number of attributes is fixed at 1.

Impact of Number of Attributes. Our method satisfies the 90% recall requirement across all feasible multi-attribute conditions

Table 6: Index size (MB).

Dataset	FAISS-IVF	IVF-RaBitQ P/I	IVF-RaBitQ PART(ours)	NHQ	FDiskANN	ACORN	SymQG InFilter	SymQG PART(ours)
SIFT	42.26	30.20	35.89	97.82	131.73	336.13	384.00	384.00
MS MARCO	281.86	161.01	169.74	99.30	131.89	336.13	384.00	384.00
OpenAI	418.38	238.44	246.03	120.12	131.85	335.79	383.62	383.62
ArXiv	1103.34	658.27	628.55	96.32	131.64	336.13	384.00	384.00

supported by the dataset. As the number of attributes increases from 1 to 3, the QPS remains highly stable, with only a marginal degradation from 3025 to 2830. This confirms that our mechanism efficiently handles multi-attribute AND predicate constraints. When the number of attributes reaches 4, the QPS drops to 453. This is not a limitation of our method; instead, it reflects the extreme boundary of the filtering space. Given that the OpenAI dataset contains $N = 999{,}000$ base vectors and the attribute cardinality is fixed at 10, the combined selectivity for four attributes decreases to 10^{-4}. This reduces the expected number of valid candidates across the entire dataset to approximately 100 ($999{,}000 \times 10^{-4} = 99.9$), which corresponds to the strictest valid query condition for a top-100 search. Notably, the index construction time remains stable under these settings, indicating that encoding additional attributes does not severely impact the offline construction.

Impact of Attribute Cardinality. The right panel shows the effect of attribute cardinality, swept from 100 to 10,000, on system performance. For low to medium cardinalities (100 to 1,000), the QPS stays consistently high (between 3,000 and 2,799), proving that the method efficiently processes standard categorical scales. As the cardinality further scales to 5,000 and 10,000, we observe a more pronounced decline in QPS (dropping to 1,846 and 375, respectively). Simultaneously, the index time exhibits a near-linear increase with larger cardinalities. This behavior is expected, as extreme high-cardinality attributes require maintaining expanded phase rotation data and more complex mapping structures, which naturally increases both the offline construction overhead and the online evaluation costs. Overall, PART effectively adapts to varying attribute cardinalities, successfully maintaining the strict 90% Recall@100 target even at a large scale of 10,000.

6.6 Validation of Theoretical Bounds

To bridge the gap between mathematical limits and real-world execution, we design a series of empirical micro-benchmarks to validate our theorems over actual data distributions. Specifically, we select three datasets with progressively increasing dimensionalities from 1024 to 4096. All vectors are normalized. For each dataset, we inject uniformly distributed categorical attributes. This methodology allows us to trace the empirical distortion between the original IP $\langle \mathbf{q}, \mathbf{o} \rangle$ and the baked IP $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle$ under varying dimensionalities.

Validation of Attribute Matching Soundness. To validate the $1/d$ convergence mechanism of Theorem 1, we track the baked IPs for mismatched vectors across the three datasets. As depicted in Figure 5, the black dashed lines represent the theoretical 3σ bounds ($y = \pm 3/\sqrt{d}$). The vertical spread of the mismatched vectors is overwhelmingly confined within these theoretical limits, achieving containment rates of 99.9%, 99.6%, and 99.9% on MS MARCO, OpenAI, and ArXiv, respectively. Moreover, this mismatch noise floor visibly narrows as the dimensionality d increases from 1024 to 4096. This

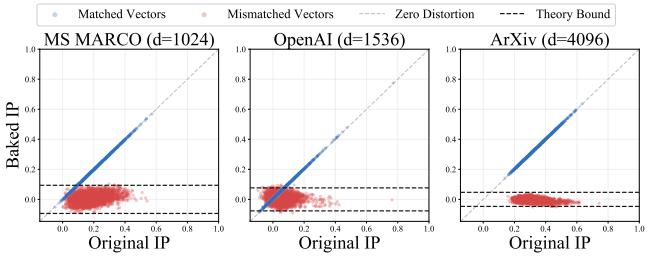


Figure 5: Verification of soundness and completeness.

alignment between the empirical spread and the theoretical limits demonstrates that PART successfully suppresses the baked IPs of attribute-mismatched vectors to near zero. It also confirms that the variance shrinkage is governed by d and inherently agnostic to the underlying data distribution. Given that vectors with near-zero IPs are virtually impossible to emerge as true top-k nearest neighbors in high-dimensional spaces, PART prunes mismatched candidates directly at the distance-evaluation level.

Validation of Vector Similarity Completeness. We analyze the baked IPs for matched vector pairs across the same datasets. As shown by the blue points in Figure 5, the baked IPs of identically attributed vectors align flawlessly along the $y = x$ diagonal. This zero-distortion phenomenon confirms that $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$ holds universally for all valid matches, regardless of the underlying vector dimensionality or data distribution. From a system perspective, this exact similarity preservation guarantees that the relative distance ranking among the filtered candidates remains entirely unperturbed. Consequently, PART supports NNS without introducing any false negatives or ranking degradation for true matches.

7 RELATED WORK

FANNS Methods. FANNS relies on execution-level strategies such as Pre-filtering, Post-filtering, or Joint-filtering (e.g., FDiskANN [13], NHQ [48], ACORN [40], and iRangeGraph [52]). Fundamentally, these strategies treat the predicate as a discrete test external to the distance computation. This separation incurs substantial inefficiency in search pruning, and no single strategy dominates across the entire selectivity spectrum [27]. Recent methods resort to selectivity-adaptive workarounds: CAPS sub-partitions each cluster by an attribute-frequency tree aimed at the "unhappy middle" regime [17], UNIFY routes Pre-/Post-/Joint-filtering within one segmented-inclusive graph [30], and FAVOR reshapes distances with a selectivity-aware exclusion term [42]. However, these designs merely re-arrange the external predicate rather than eliminating it. By baking attributes into the high-dimensional vector space, PART resolves the selectivity dilemma at its root, achieving high query throughput and recall stability across all selectivity regimes.

Beyond the fundamental challenge of selectivity, recent studies also emphasize dynamic maintenance and versatile arbitrary filtering. Methods that physically fuse predicates into the index topology (e.g., SeRF [58], iRangeGraph [52]) require costly structural reconstruction upon attribute updates and are restricted to a single predicate type. While dynamic range indices like DIGRA [21], RangePQ [55], and WoW [50] support incremental updates, they rely on auxiliary structures. In contrast, PART seamlessly operates on general-purpose indices via the RaBitQ-PART quantization framework, enabling simultaneous attribute matching and vector distance estimation with strict error guarantees, thereby rapidly pruning invalid candidates and effectively improving search efficiency. When attribute updates occur, PART only requires a rapid re-encoding of the quantized codes, completely eliminating the need for costly index reconstruction. Furthermore, it natively accommodates commonly used complex constraints, including multi-attribute AND, numerical range filtering, and hybrid predicates.

Furthermore, existing methods struggle with skewed attribute distributions. For instance, attributes in the ArXiv dataset [20] exhibit heavy-tailed, power-law-like cardinalities. Under such non-uniform data, heuristic joint-filtering methods lack hard correctness guarantees and suffer unexpected recall collapses even after careful parameter tuning. Current mitigations remain highly distribution-specific, such as graph re-balancing in TFANNS [33] or dedicated auto-tuning via PGTuner [8], and fail to generalize to unseen skewed distributions. In contrast, PART embeds constraints directly into the vector coordinates via high-dimensional angular rotations. Backed by strong theoretical guarantees, PART ensures that massive discrepancies in attribute cardinalities do not degrade the accuracy of vector search or attribute matching. Consequently, it naturally accommodates heavily skewed data without requiring distribution-specific index structures or auto-tuning.

Rotation-Based Encoding. The primitive underlying PART which encoding a discrete signal as a phase rotation evaluated via a plain IP is well-established in adjacent fields. For instance, RoPE [43] represents token positions as complex-plane rotations, RotatE [45] models knowledge-graph relations as rotational operations. Rotations are also prevalent in VQ. OPQ aligns vector spaces with quantization subspaces [35], and RaBitQ [12] applies a random projection prior to one-bit sign quantization to achieve a distribution-independent $O(1/\sqrt{d})$ error bound. PART builds directly on the RaBitQ framework by inserting an attribute-driven rotation *between* RaBitQ's preconditioning and its sign quantization. In doing so, PART repurposes the orthogonal transformation for double duty: it simultaneously decorrelates coordinates for quantization and embeds the filtering semantics.

8 CONCLUSION

In this paper, we proposed PART, a novel method that resolves the low selectivity dilemma in FANNS. By baking discrete attribute constraints into high-dimensional vectors via orthogonal phase rotations, PART translates predicate matching into vector distance computation with robust theoretical error guarantees. This design can be naturally extended to support complex query conditions, including multi-attribute AND semantics, numerical range filtering, and hybrid predicates. Furthermore, it serves as a versatile drop-in frontend that can be readily integrated with mainstream ANNS architectures such as VQ, IVF, and proximity graphs. Extensive evaluations across diverse workloads, including heavily skewed real-world datasets, demonstrate that PART achieves state-of-the-art query throughput across the entire selectivity spectrum while robustly maintaining strict recall targets.

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